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Estimation of Carbon Balance in Steppe Ecosystems of Russia

L. L. Golubyatnikov[a,] *, I. N. Kurganova[b], and V. O. Lopes de Gerenyu[b]
[a] *Obukhov Institute of Atmospheric Physics, Moscow, 119017 Russia*
[b] *Institute of Physicochemical and Biological Problems in Soil Science, Pushchino, Moscow oblast, 142290 Russia*
**e-mail: golub@ifaran.ru*
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Abstract—Steppe ecosystems, occupying about 8% of the terrestrial area, are an essential element of the global carbon cycle in the atmosphere–vegetation–soil system. The carbon (C–CO_2) balance of natural steppe ecosystems in Russia is estimated based on the geoinformation–analytical method and employing the database of empirically measured values of the net primary production and a climate-driven regression model that makes it possible to estimate the intensity of carbon dioxide flux from soils into the atmosphere. Natural steppes in Russia serve as a significant sink of carbon dioxide from the atmosphere. The average intensity of this carbon flux can be estimated at 231 ± 202 gC/m^2 per year. The estimated annual accumulation of carbon dioxide in the natural steppe ecosystems of Russia is 111 ± 97 MtC. According to the estimates, the steppe ecosystems under study provide from 8 to 19% of the atmospheric carbon sink to the terrestrial ecosystems of Russia.

Keywords: atmospheric carbon update, carbon dioxide emission, C–CO_2 balance, steppe ecosystems
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INTRODUCTION

Terrestrial ecosystems play a pivotal role in the global carbon turnover and have a considerable influence on the concentration of greenhouse gases in the atmosphere. Vegetation cover of terrestrial ecosystems annually assimilates 150–175 GtC of atmospheric carbon dioxide (CO_2) resulting from photosynthesis [1–3], contributing to the gross primary production (*GPP*) of vegetation. About 50–60% of plant-absorbed carbon returns to the atmosphere as a result of the autotrophic respiration (R_A) of plants [4, 5]. An indicator of intensity of atmospheric carbon accumulation by vegetation is net primary production (*NPP*), which is determined by the amount of organic matter accumulated for some specific time interval (usually a year) per unit of area in the aboveground and belowground parts of a vegetation community [6]; i.e., $NPP = GPP - R_A$. According to [7, 8], the vegetation of terrestrial ecosystems annually accumulates 29–37% of CO_2 that enters the atmosphere because of human activities. The CO_2 flux from terrestrial ecosystems into the atmosphere, forming as a result of the microbial decay of soil organic matter and plant detritus, is referred to as soil heterotrophic respiration (R_H). Soil heterotrophic respiration and root respiration are the main natural sources of CO_2 transferred from the soil surface into the atmosphere. This flux is referred to as soil respiration (R_S) [9, 10]. Based on the data from [11–15], estimated R_S for terrestrial ecosystems are within the range of 68–100 GtC/yr. Soil respiration is the most intensive CO_2 flux from the terrestrial ecosystems into the atmosphere; therefore, its smallest variation on the global scale can lead to a considerable change in the concentration of atmospheric CO_2 [12, 16]. According to estimates [12, 14, 17, 18], global soil respiration presently exceeds by an order the amount of anthropogenic CO_2 discharges produced by the combustion of fossil fuels.

The functioning of an ecosystem as a sink or source of atmospheric CO_2 is determined by the sign of carbon balance (C–CO_2) value for the considered ecosystem. The value of carbon balance value for a natural ecosystem, also referred to as net ecosystem production (NEP), is equal to the difference between carbon input into an ecosystem (as a result of production processes) and carbon emission (in the form of CO_2, as a result of destruction of the organic matter accumulated in an ecosystem) [16, 19–21]. A considerable uncertainty in estimates of the global carbon budget is related to the different degree of studying the intensity of carbon fluxes at the level of both individual ecosystems and entire regions.

The steppe is one of the natural zones for which the elements of the carbon cycle are quite poorly studied. It is an arid ecosystem where the vegetation cover is formed mainly by various grass species with the predominance of perennial grasses. Steppe ecosystems occupy 8% of ice-free land area and represent a significant carbon reservoir because of the high organic matter content in soils: within a 1-m layer, organic matter content is 22.9–35.7 kgC/m^2 [22–26]. The steppe zone is constituted by Eurasian steppes, North American prairies, prairielike graminoid communities of Southeast Australia, South American pampas, South African veldts, and the tussocks of New Zealand [27]. The largest steppe areas are located in Eurasia. Eurasian steppes occupy about 8000000 km^2 and form a latitudinal band of 150 to 600 km wide [28], extending from the Pannonian Plain in the west to the Northeast China Plain in the east (between 41°–57° N and 26°–127° E).

The steppe ecosystems are characteristic of a broad variation range of climatic characteristics which determine the zonality and degree of contrast of the soil and vegetation cover of the steppe zone [27, 29]. According to Köppen's climate classification [30], the steppe zone is characterized by the steppe, desert, warm marine, moderate marine, warm continental, and moderate continental climate types.

The distinctive peculiarity of the steppe vegetation is its extensive root system; the fraction of belowground phytomass can be up to 87% of the total phytomass [31]. The steppe ecosystems are where the most fertile soil types are developed: chernozems and chestnut soils in Eurasian steppes and chernozem-like soils in North American prairies and South American pampas. The steppe zone is the main cropping region of the world and is characterized by a high degree of anthropogenic degradation of the natural ecosystems. A considerable part of the steppe areas is plowed and used to cultivate crops. Large steppe areas are used as pastures and hayfields [29].

Soil and vegetation cover, climatic conditions, and intensity of anthropogenic influence determine the main parameters of carbon cycle in steppe ecosystems and the ability of these ecosystems to accumulate and emit greenhouse gases. Among all types of terrestrial ecosystems in the territory of Russia, the role of steppe ecosystems in the biogenic carbon cycle is the least defined. In the present work, the estimates of carbon balance ($C-CO_2$) and its components (NPP, R_H) are obtained for steppe ecosystems of Russia and the contribution of steppe ecosystems into the carbon balance of Russia is estimated.

STUDY OBJECTS

According to the map of vegetation [32], steppe ecosystems occupy about 1700000 km^2 (i.e., less than 10%) of the territory of Russian Federation. Within the Russian territory, the steppe zone covers vast regions in the southern East European Plain, plain areas of Fore-Caucasus and Crimean regions, the southern part of the West Siberian Plain, and intermontane depressions of Central Asia (the Republic of Buryatia, Republic of Khakassia, Republic of Tyva, Irkutsk oblast, Zabaikalsky krai, and the southern part of Krasnoyarsk krai) (Fig. 1). The steppe ecosystems of Russia are subdivided into meadow, true, dry, and desert. The subzone of meadow steppes stretches along the southern boundary of the forest zone. In the territory of Russia, meadow steppes form the largest in area steppe subzone (710000 km^2). The ecosystems of true steppes in Russia cover about 470000 km^2. These ecosystems occupy considerable areas in the southwestern European part of Russia, while occur sporadically in the Asian part. Dry steppes are distributed predominantly in administrative regions such as Stavropol krai, the Republic of Crimea, the Republic of Dagestan, and the Republic of Tyva. The area of this subzone in Russia is 320000 km^2. Ecosystems of desert steppes in Russia cover about 230000 km^2, occupying significant areas in the southeastern European part of Russia and in particular intermontane depressions of central and southern Tyva.

The steppe ecosystems demonstrate the soil moisture deficit, which increases from meadow to desert steppes. The largest part of precipitation (75–85%) within the steppes of Russia occurs in the warm season. The annual precipitation ranges from 430 mm in meadow steppes to 150 mm in desert ones. Free-water-surface evaporation varies from 650 mm/yr in meadow steppes to 800 mm/yr in desert ones. In the steppe regions of Russia, cumulative annual near-surface air temperatures above 10°C fall within the range from 1900 to 2600°C [27].

A considerable part of Russian steppes is occupied by agricultural lands (arable lands, pastures, and hayfields) [33]. The areas of 42 administrative regions of Russian Federation are fully or partially located in the steppe zone. Natural steppe ecosystems remained within relatively small areas. According to the land inventory of the remained steppe areas [34], the total area of natural steppe ecosystems in Russia is estimated at 200000 km^2, comprising less than 12% of the total steppe-covered areas of Russian Federation.

In the 1990s, due to the dramatic socioeconomic changes in the country, significant areas of arable lands were withdrawn from agricultural activities. Noncultivated arable lands have been withdrawn to the pool of abandoned lands. Due to the reduction in the numbers of livestock, the anthropogenic load on pastures decreased, as did the area of pastures and hayfields. As a result of these events, the restoration of natural steppe ecosystems began in some agricultural lands of the steppe zone. Under favorable restoration conditions, it took 15–30 years for the grass stratum to form in these ecosystems; the species composition of this cover is similar to steppe plant communities that existed before plowing [35, 36]. When forming the vegetation cover of steppes, the amount of organic matter supplied into the soil increased and the natural physicochemical properties and microbiological characteristics of soils in former agricultural lands gradually restored [37]. In particular, soil horizons of abandoned lands demonstrate the changes is structure,

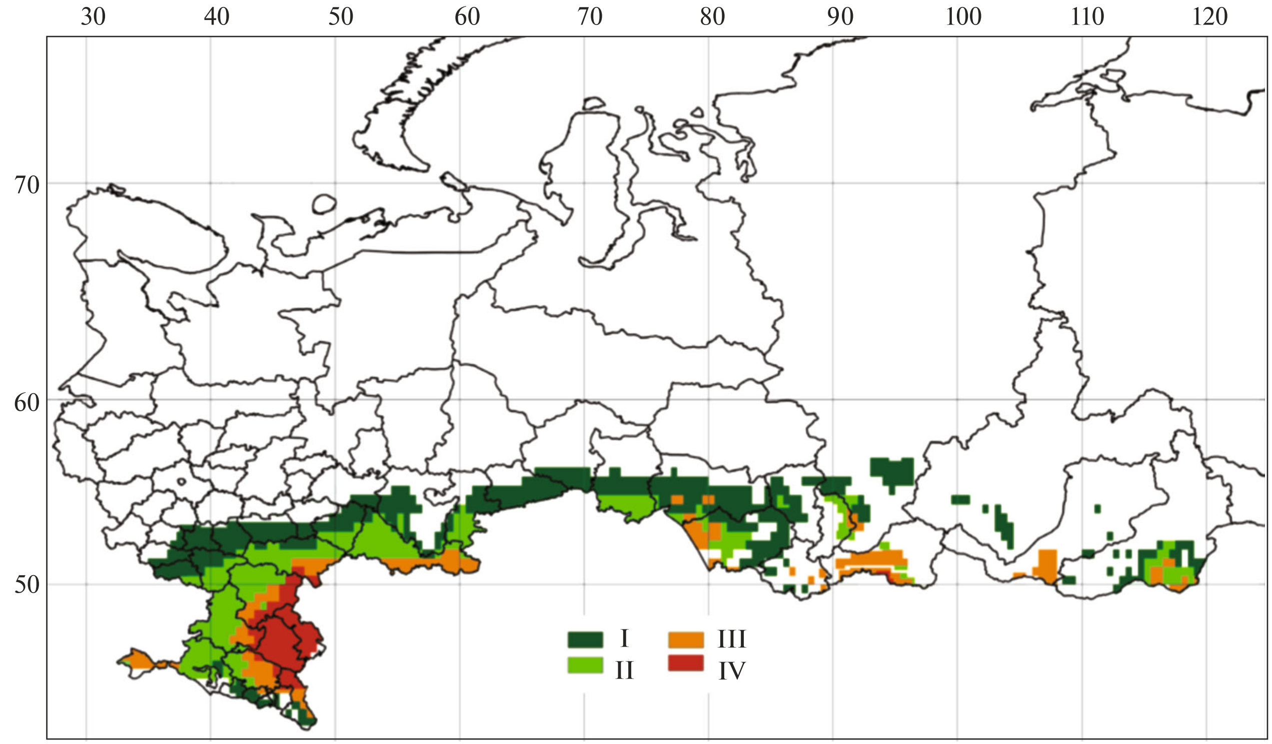

Fig. 1. Schematic map showing the distribution of different types of steppes within the territory of Russia: I, meadow; II, true; III, dry; and IV, desert.

density, air-and-water and hydrothermal regimes, and the increase in contents of carbon and mineral nutrients for plants [38–40]. In present work we consider the natural and restoring (secondary) steppe ecosystems. Hereinafter, we will refer to these ecosystems as natural ecosystems of the steppe zone.

In the growing period, the intensity of photosynthesis-driven atmospheric carbon accumulation in the natural steppe ecosystems exceeds the intensity of carbon release into the atmosphere due to destruction processes [16, 41]. In the cold season, as a result of (a) flagging and dying-off of aerial parts of plants in the grass stratum and (b) the formation of the snow cover, the absorption of atmospheric carbon by the vegetation of steppe ecosystems ceases, whereas the microbial decay of soil detritus and respiration of belowground parts of plants continues (i.e., CO_2 flux from the soils of steppe ecosystems into the atmosphere goes on). It should be noted that, in the cold season, due to low air temperatures and the freezing of the upper soil layers, the intensity of destruction processes in the steppe ecosystems considerably decelerates compared to the warm period. Thus, in the warm season, natural steppe ecosystems represent a sink of atmospheric carbon, whereas in the cold season they act as a source of CO_2 supply into the atmosphere. Plant-consumed atmospheric carbon accumulates mainly in organic matter of the soil pool of ecosystems (root system, phytodetritus, and humus) [14, 42]. According to the data from [31, 36, 42, 43], belowground parts of plants in the steppe ecosystems contain as much as 80–95% of carbon from the total phytomass.

METHODS

Areas of Secondary Steppe Ecosystems

Areas of secondary steppe ecosystems were calculated on the basis of the data from (a) statistical reports on the availability and distribution of lands in administrative regions of Russian Federation as of January 1, 2020 (prepared by Federal Service of State Registration, Cadastre, and Cartography, briefly Rosreestr), and (b) bulleting on the state of the agriculture in Russian Federation in 2020 (issued by the Federal Service of State Statistics, briefly Rosstat). The Rosreestr statistical data on lands [44] considers the agricultural lands that were officially withdrawn from arable to abandoned lands in administrative regions of Russian Federation. However, a certain part of noncultivated arable lands in administrative regions of Russia was not withdrawn to other categories of agricultural lands and is still formally considered croplands. In this respect, the real areas of abandoned lands were estimated on the basis of the Rosreestr data on areas of arable lands and declared abandoned lands [44] taking into account the annual Rosstat reports on areas under crops and areas of complete fallow fields [45]. The restoration of steppe ecosystems also takes place in the areas that ceased to be used as pastures and hayfields. The areas of secondary steppes developed after pastures and hayfields was made on the basis of (a) Rosreestr statistical data on the areas of these categories of agricultural lands [44] and (b) the data on areas of pastures and hayfields in use by enterprises and citizens from Rosstat annual reports [45].

In a number of the considered administrative regions of Russian Federation, abandoned croplands, unused pastures, and unused hayfields are located outside the steppe zone as well. There are no data on the areas of agricultural lands where anthropogenic influence ceased and which are located beyond the steppe zone. As an upper estimate for the areas of these lands, I. Smelyanskii [35] proposed as many as 20% of the area that ceased to be used for agricultural purposes in any given administrative region of Russia.

Estimation of Net Primary Production

Estimated values of net primary production of the natural steppe ecosystems in different administrative regions of Russian Federation were obtained on the basis of geoinformational–analytical method using the database of empirically measured values of this parameter. In our present work we used the data on NPP of steppe ecosystems from [28, 31, 32, 36, 43, 46–52]. To convert the NPP values expressed in dry matter units into carbon units, we used the coefficients obtained on the basis of analytical data on nitrogen content (N) and C/N ratio in phytomass of the considered (steppe) ecosystems [53].

Estimation of Soil Respiration

The values of soil respiration were estimated on the basis of regression T&P model [54]. This model makes it possible to estimate the mean monthly intensity of CO_2 release from soils (R_{Sd}, gC/(m^2 day)) on the basis of mean monthly near-surface air temperature (T, °C) and monthly amount of precipitation (P, cm/month):

$$R_{Sd} = Fe^{QT} P/(K + P), \qquad (1)$$

where F is the soil respiration (gC/(m^2 day)) at 0°C and sufficient moistening of soil; Q (°1/C) is the coefficient of temperature sensitivity of soil respiration; and K (cm/month) is the half-saturation constant of the hyperbolic relationship between R_{Sd} and P. A calculation of the parameters for the T&P model was performed by the nonlinear least-squares method in the R software environment for statistical processing of data (nlm function) [55]. Monthly fluxes of CO_2 from soils of steppe ecosystems were obtained by multiplying R_{Sd} values to the number of days of the respective month. The annual flux of CO_2 from soils of ecosystems (R_S) was calculated by the summation of computed monthly values of soil respiration. The soil respiration of steppe ecosystems was calculated on the basis of meteorological data for the period from 1970 to 2000, stored in the WorldClim database (www.worldclim.org).

On the basis of the experimental data for different natural climatic zones, it has been found that the fraction of heterotrophic respiration in the total flux of CO_2 from soils of grass ecosystems is 55% [56]; i.e., R_H and R_S are reliably described by the following relation:

$$R_H = 0.55R_S. \qquad (2)$$

Carbon Balance

We calculated the carbon balance (C–CO_2) of the natural steppe ecosystems by the following formula:

$$NEP = NPP - R_H. \qquad (3)$$

If the NEP value of an ecosystem is positive, then this ecosystem is a sink of atmospheric carbon. On the other hand, if the NEP value is negative, then the ecosystem is a source of atmospheric carbon.

To estimate the carbon balance of the natural steppe ecosystems in warm and cold seasons, we assumed that the warm season included the months when $T \geq +1$°C and the cold season included the months when $T < +1$°C. The durations of warm and cold seasons in steppe zone were determined separately for each of the administrative regions of Russian Federation under consideration.

The statistical data on lands (from Rosreestr and Rosstat) were clustered on administrative regions of Russian Federation. In this respect, the steppe ecosystems and the values of NPP, R_H, and NEP fluxes for these ecosystems were calculated for each individual administrative region of Russia where steppes are present. The calculations were performed on a grid with spatial resolution of 0.5 × 0.5 degrees. On the basis of the results, we determined mean values of the NPP, R_H, and NEP for the natural ecosystems for Russian Federation in general and for its European, Siberian, and Central Asian steppe regions in particular. Calculations of mean values of the fluxes were made taking into account the areas occupied by the natural steppe ecosystems in every administrative region of Russian Federation.

RESULTS AND DISCUSSION

Average Annual Near-Surface Air Temperatures and Annual Atmospheric Precipitation

The Russian territories under steppes are characterized by considerable variations of climatic parameters in both latitudinal and longitudinal directions. It should be noted that the variation ranges of climatic parameters in the steppe regions of Russia are considerably larger than the analogous ranges of the mid-latitude forest regions, although forests occupy larger areas in Russia than steppes [27].

Table 1. Statistical characteristics of average annual near-surface air temperature (T_{yr}) and annual precipitation (P_{yr}) for the steppe regions of Russia

Steppe region	T_{yr}, °C						P_{yr}, mm/yr					
	x_{min}	x_{max}	A	$\bar{x}$	σ	Cv, %	x_{min}	x_{max}	A	$\bar{x}$	σ	Cv, %
European	2.5	10.9	8.4	7.3	2.5	34	237	769	532	420	121	29
Siberian	–2.9	2.1	5.1	0.9	1.6	184	395	578	183	444	48	11
Central Asian	–5.0	–0.6	4.4	–3.6	1.8	50	372	583	211	437	71	16
Steppe regions of Russia in general	–5.0	10.9	15.9	3.1	5.2	170	237	769	532	430	98	23

x_{max}, the maximum value; x_{min}, minimum value; A, variation amplitude; $\bar{x}$, mean value; σ, standard deviation; Cv, variation coefficient.

According to the meteorological data, the values of average annual near-surface air temperature (T_{yr}) for steppe areas vary between administrative regions of Russian Federation: it ranges from –5.0 to –4.8°C in republics of Buryatia and Tyva but from 10.5 to 10.9°C in Krasnodar krai and Republic of Crimea. Annual precipitation (P_{yr}) in the steppe regions of Russia varies from 237–284 mm/yr in Astrakhan oblast and Republic of Kalmykia to 765–769 mm/yr in Krasnodar krai and the Karachay-Cherkes Republic. The greatest variation range between T_{yr} (8.4°C) and P_{yr} (532 mm/yr) is observed in the steppe ecosystems of the European steppe region of Russia. The statistical characteristics of T_{yr} and P_{yr} values for steppe regions of Russian Federation are presented in Table 1.

Average annual near-surface air temperatures for the warm/cold seasons in the steppe ecosystems of Russia change from 10.2°C/–13.4°C in the Central Asian steppe region to 14.8°C/–4.7°C in the European one (Fig. 2). Atmospheric precipitation of the warm season in the steppe regions of Russia exceeds that of the cold season 2.2–3.4 times. The Central Asian steppe region of Russia is characterized by the highest precipitation in the warm season (338 mm/season, Fig. 2a), while the lowest one in the cold season (99 mm/season, Fig. 2b).

Areas of Natural and Secondary Steppe Ecosystems

In [34], on the basis of the field studies and analysis of medium-resolution satellite images, relatively large areas of natural steppe ecosystems were revealed in the territory of Russia. The largest part (131000 km^2) of the remaining steppe ecosystems is located in the European steppe region. In the Siberian and Central Asian steppe regions, the areas of remaining steppes are 18000 and 53000 km^2, respectively (Fig. 3a).

After the use of certain agricultural lands in the steppe zone stopped, the restoration of steppe ecosystems had begun in the areas formerly occupied by arable lands, pastures, and hayfields. According to the calculations, by early 2020, secondary steppe ecosystems in Russia occupied 277000 km^2; i.e., their area exceeded the area of natural steppes by 38%.

The largest part (57%) of secondary steppe ecosystems is located in the Asian part of Russia. The areas of steppe ecosystems, which developed after croplands, pastures, and hayfields were abandoned, are virtually equal in the Asian and European parts of Russia, making up some 80000 km^2 (Fig. 3a). The areas occupied by secondary steppe ecosystems, which developed after abandoned pastures and hayfields, are larger in the Asian part of Russia: twice as much as in the European part (Fig. 3b).

Identification of T&P Model

Multiannual instrumental studies of the intraannual dynamics of CO_2 emission from soils of natural steppe ecosystems of Russia have not been carried out. In this respect, we identified T&P model (1) on the basis of the data from multiannual field measurements of CO_2 fluxes into the atmosphere from soils of two meadow ecosystems located in the southern Moscow oblast. Weekly measurements of CO_2 emission from soils had been being performed continuously at two sites: (1) since 1998 until 2017 by the closed chamber method at the grass–forb meadow of the Priioksko-Terrasnyi Biospheric Reserve and (2) from 2004 to 2018 at forb–grass meadow near the town of Pushchino. The respective measurements recorded the CO_2 flux into the atmosphere from meadow soils; this flux included CO_2 that formed as a result of root respiration, the microbial decay of aboveground detritus and microbial decay of soil organic matter. Meadow ecosystems differed in species compositions of their vegetation, as well as in soil types and structures. The considered meadow ecosystems are characteristic of the considerable species diversity of grass stratum, thick sod formed by gramineous plant roots, and the values of phytomass and NPP being close to the analogous parameters of meadow steppes. Hence, with certain reservations, the studied meadow ecosystems can be considered as close to the ecosystems of meadow steppes. In our calculations we used the

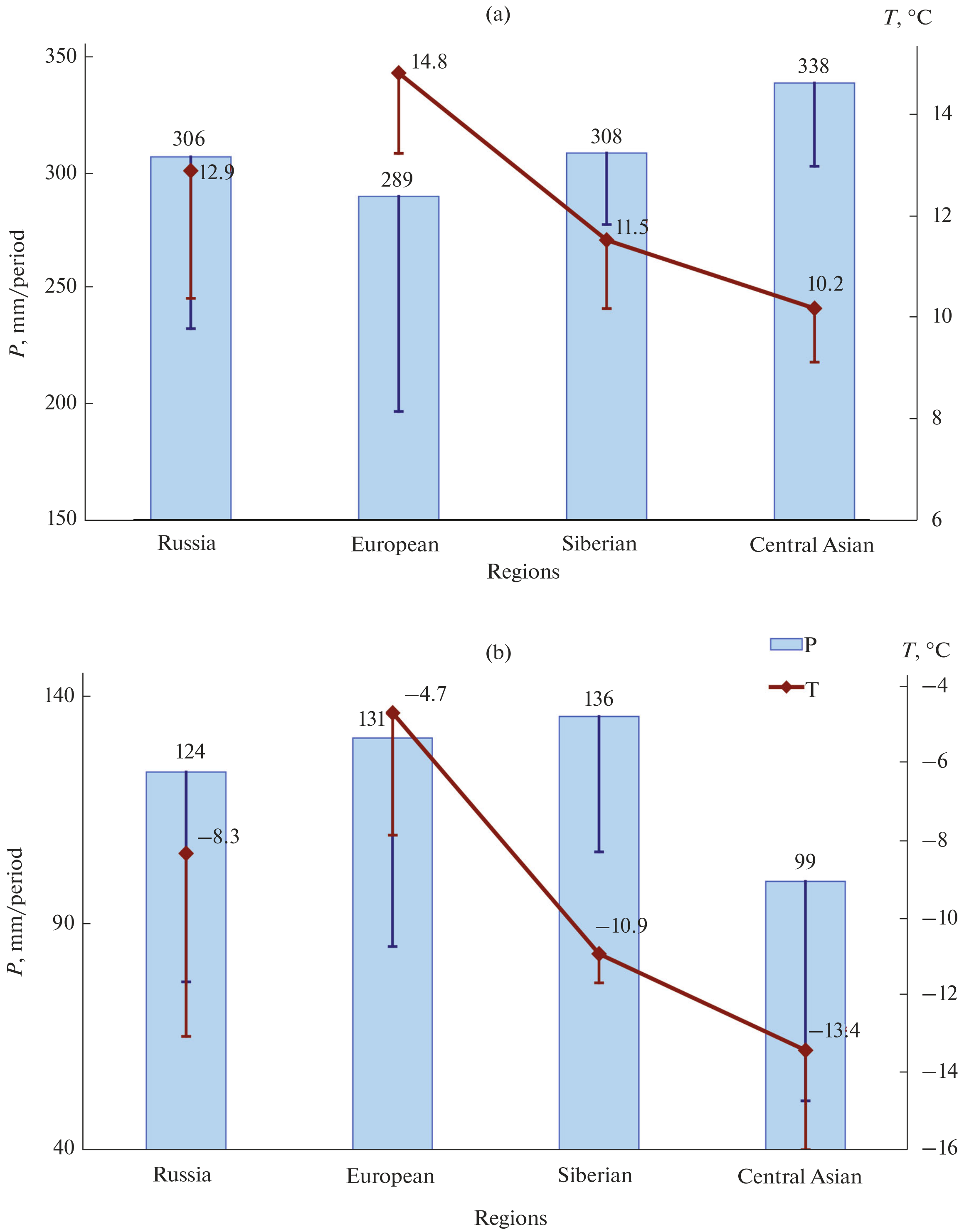

Fig. 2. Near-surface air-temperature (T) and precipitation (P) in steppe regions of Russia (mean values for the period and standard deviations): (a) warm season and (b) cold season. Vertical lines denote negative components of standard deviations.

meteorological data from the Background Monitoring Station (settlement of Danki, Serpukhov district, Moscow oblast), which is located at one of the sites where experimental studies of soil respiration were carried out.

On the basis of the experimental data for each meadow ecosystem under consideration, we determined model parameters F, Q, and K (Table 2). The computations showed that the two obtained T&P models (T&P-G1 and T&P-G2) adequately describe

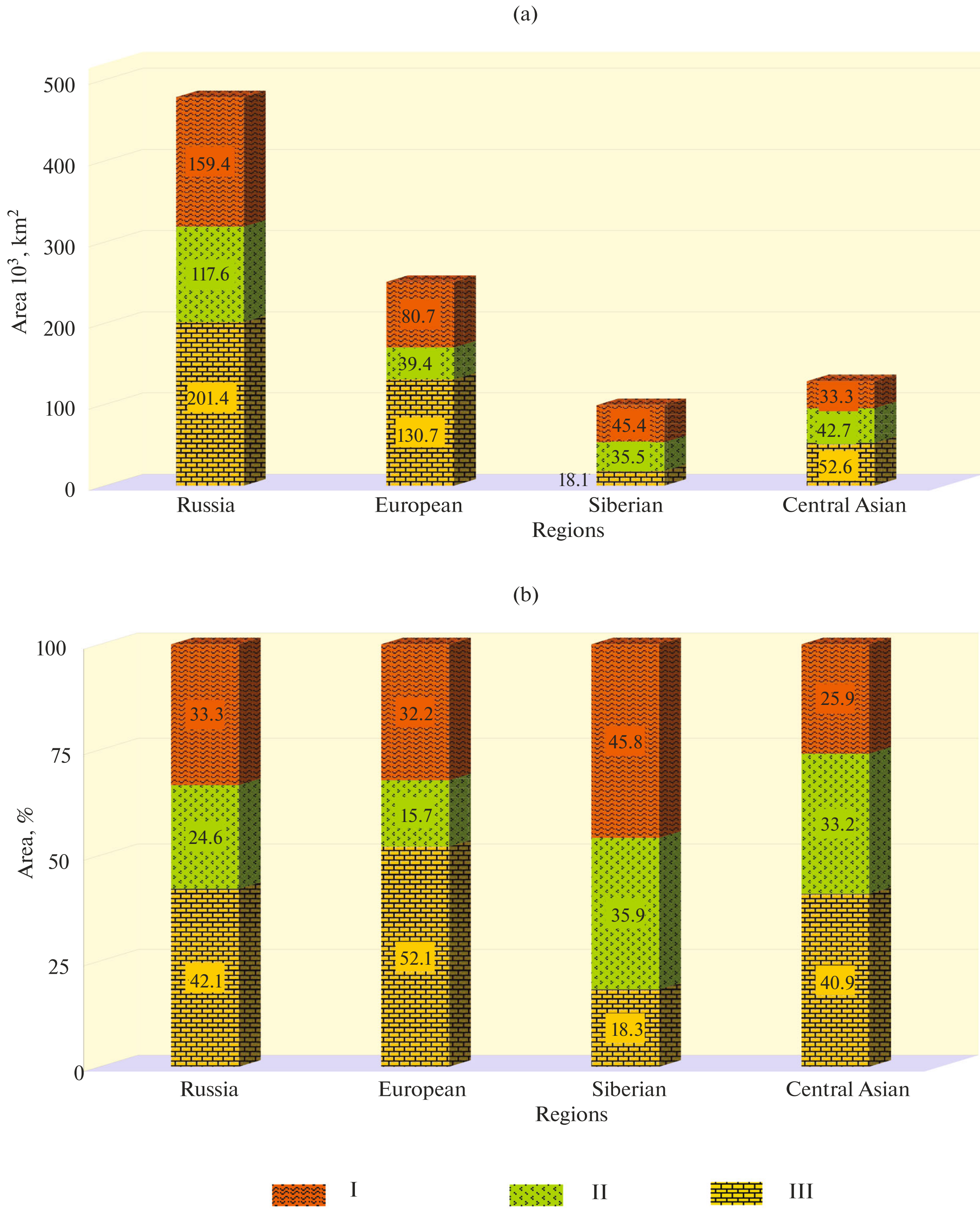

Fig. 3. Distribution of the areas of natural and secondary steppe ecosystems in steppe regions of Russia and in steppes of Russia on the whole. Steppe ecosystems: I, upon abandoned croplands; II, upon abandoned pastures and hayfields; and III, natural.

the intraannual dynamics of the mean monthly CO_2 emission from the soils of the studied ecosystems (Table 3). Theil's inequality coefficients (U) for the series of mean monthly soil respiration values and the data of instrumental measurements of this value do not exceed 0.3 (Table 2), which is the threshold value for natural processes [57, 58]. The values of coefficient U argue for the fact that the modeling results have a great degree of correspondence to the instrumentally measured data. The values of determination coefficient R^2 (Table 2) also indicate a high correlation between the computed and experimental values of

Table 2. Parameters and estimated accuracy of T&P models for the calculation of C–CO_2 fluxes from soils of meadow ecosystems

Study object	Model parameters				Estimated accuracy			
	Model	F, gC/(m^2 day)	Q, ° 1/C	K, cm/month	R^2		U	
					T&P-G1	T&P-G2	T&P-G1	T&P-G2
Ecosystem 1	T&P-G1	1.687	0.0569	2.203	0.62	0.64	0.19	0.24
Ecosystem 2	T&P-G2	1.843	0.0654	2.745	0.71	0.70	0.20	0.17

R^2 is the determination coefficient; U is Theil's inequality coefficient.

mean monthly CO_2 fluxes into the atmosphere from the soils of the considered ecosystems.

Based on the model and experimental data on monthly CO_2 fluxes into the atmosphere from soils of the considered meadow ecosystems, we estimated the annual values of R_S for these ecosystems. The results showed that the deviation of the modeled annual values of R_S from the experimental estimates of this parameter in meadow ecosystems falls within the range of 1 to 26% depending on the T&P model. However, the deviation of the mean values of annual R_S, which were obtained on the basis of model estimates of this flux in each meadow ecosystem, from the annual experimental values of parameter R_S is 6–11%. In the present study, the value of annual CO_2 flux into the atmosphere from soils of steppe ecosystems was calculated as the mean value of the respective annual CO_2 fluxes computed from the data of the T&P-G1 and T&P-G2 models.

Table 3. Mean values ($\overline{x}$) and standard deviations (σ) of the experimental (FieldData) and model (T&P-G1, T&P-G2) values of soil respiration (gC/(m^2 day)) for two meadow ecosystems

Month	Ecosystem 1				Ecosystem 2			
	FieldData		T&P-G1		FieldData		T&P-G2	
	$\overline{x}$	σ	$\overline{x}$	σ	$\overline{x}$	σ	$\overline{x}$	σ
January	0.66	0.41	0.72	0.20	0.63	0.33	0.67	0.24
February	0.74	0.48	0.73	0.22	0.51	0.27	0.67	0.21
March	0.74	0.42	0.97	0.21	0.55	0.26	0.97	0.26
April	1.34	0.57	1.50	0.35	1.46	0.58	1.57	0.47
May	2.69	0.88	2.46	0.53	3.32	0.89	3.05	0.53
June	3.54	1.23	3.18	0.43	4.73	1.59	3.78	0.63
July	3.86	1.52	3.77	0.60	4.45	1.64	4.40	0.80
August	3.00	1.44	3.19	0.88	3.55	1.46	4.04	1.01
September	2.44	0.88	2.26	0.38	2.82	1.03	2.46	0.52
October	1.61	0.65	1.62	0.33	1.69	0.65	1.73	0.45
December	0.96	0.47	1.05	0.22	1.13	0.34	1.10	0.26

Atmospheric Carbon Update by Vegetation of Steppe Ecosystems

The NPP value for steppe ecosystems decreases from meadow to desert steppes, obeying the aridity factor [31, 47]. For the meadow steppes of Russia, NPP is 755 ± 64 gC/(m^2 yr) (mean ± standard deviation). In the considered desert steppes, the NPP value drops to 311 ± 36 gC/(m^2 yr). The mean NPP value for the natural steppe ecosystems of Russia is 594 ± 148 gC/(m^2 yr). It should be noted that the obtained mean NPP value of the studied steppe ecosystems is 1.9 times higher than the mean NPP value for the forest ecosystems of Russia, which is 319 ± 19 gC/(m^2 yr) [59].

According to the data, the NPP values for natural steppes of the European region change from 282 to 779 gC/(m^2 yr), with the mean value being equal to 494 gC/(m^2 yr). The NPP values of steppe ecosystems of the Siberian region vary from 656 to 790 gC/(m^2 yr), and the mean value is 730 gC/(m^2 yr). NPP of vegetation communities from intermontane depressions of the Central Asian steppe region of Russia range from 435 to 741 gC/(m^2 yr), and the mean value is 613 gC/(m^2 yr). The statistical characteristics of NPP values for the considered steppe regions of Russia are given in Table 4. The estimates show that productivity of natural steppes in the European steppe region of Russia insignificantly exceed that of forests (454 gC/(m^2 yr) [59]) in the European part of Russia. The mean productivity of steppe ecosystems in the Siberian steppe region is 1.5–1.8 times higher than productivity of forest (southern taiga) ecosystems of Central Siberia (405–475 gC/(m^2 yr) [60]).

According to the results, the vegetation cover of the natural steppe ecosystems in the European region of Russia shows carbon accumulation rate of 124 ± 5 MtC/yr; Central Asian region, 79 ± 6 MtC/yr; and Siberian region, 72 ± 4 MtC/yr. The estimates of atmospheric carbon accumulation rate by the natural steppe ecosystems of the mentioned steppe regions of Russia (relative to the total NPP of steppes in Russia) are 45, 29, and 26%, respectively.

C–CO_2 Effluxe into the Atmosphere from Soils of Steppe Ecosystems

According to the computations, the mean annual flux of carbon dioxide from soils of the steppe zone within the territory of Russia is 624 ± 142 gC/(m^2 yr). The estimate of the annual soil respiration of steppe ecosystems is comparable to R_S values (319–735 gC/(m^2 yr) [61–63]), which were reported in forest ecosystems of moderate latitudes of Russia. A few empirical estimates of CO_2 effluxes into the atmosphere from steppe soils of the Russian territory are based on the data of several measurements carried out within one or two growing seasons. According to the published data of instrumental measurements, the CO_2 fluxes into the atmosphere from soils of particular natural steppe ecosystems of Russia were recorded in the range from 194 to 3269 gC/(m^2 yr) [64, 65]. The considerable scatter of R_S values for steppe ecosystems is determined, first and foremost, by the diversity of soil and vegetation covers of steppes, by weather conditions during the measurements, and by the differences in methods that were used to measure soil respiration [66].

The published estimates of R_H contribution into CO_2 emission from soil of grass ecosystems exceed 50%. According to L. Mukhortova et al. [67], the model estimates of the fraction of heterotrophic respiration in soil respiration of steppe ecosystems of Russia change from 59 to 71%, with the mean value of 68%. It is noted in [68, 69] that North American prairies have R_H/R_S = 60–75%. In [70], the contribution of R_H to soil respiration of grass ecosystems of Central Italy is estimated at 64–72%. The statistical estimate of R_H fraction in soil respiration of grass ecosystems (55%) used in the present work is close to the lower boundary of the variation range of this value for grass ecosystems of middle latitudes (58–72%) [71] and is consistent with the analogous estimate (52%) obtained on the basis of instrumental measurements at meadow and steppe sites of the Central Chernozem Biospheric Reserve (Kursk oblast) [72].

According to the estimates, soil heterotrophic respiration of the natural steppes in the European region of Russia changes from 348 to 554 gC/(m^2 yr) with the mean value of 405 gC/(m^2 yr). Steppe ecosystems of the Siberian region of Russia have R_H varying from 240 to 337 gC/(m^2 yr) with the mean value of 312 gC/(m^2 yr). Steppe ecosystems of intermontane depression of the Central Asian region of Russia are characterized by R_H values from 216 to 296 gC/(m^2 yr), and the mean value is equal to 247 gC/(m^2 yr). The mean value of R_H for the natural steppe ecosystems within the territory of Russia is estimated at 343 ± 78 gC/(m^2 yr). Statistical characteristics of R_H values of the considered steppe regions are given in Table 4. The model estimates of heterotrophic respiration of the natural steppe ecosystems exceed the analogous estimates for the steppes of the European part of Russia (250–285 gC/(m^2 yr)) and are comparable with the estimates for steppes of the Asian part of Russia (219–246 gC/(m^2 yr) [67]). The estimate of the mean R_H value for the natural steppe ecosystems of Russia is consistent with the estimate of the same value from [73]: 390 gC/(m^2 yr).

Table 4. Statistical characteristics of net primary production, soil heterotrophic respiration, and carbon balance in steppe regions of Russia

Steppe region	x_{min}	x_{max}	A	$\bar{x}$	σ	Cv, %
	Net primary production					
European	282	779	497	494	160	32
Siberian	656	790	134	730	39	5
Central Asian	435	741	306	613	108	18
Steppe regions of Russia on the whole	282	790	509	575	160	28
	Soil heterotrophic respiration, R_H					
European	348	554	206	405	43	11
Siberian	240	337	98	312	30	10
Central Asian	216	296	79	247	26	11
Steppe regions of Russia on the whole	216	554	338	343	78	23
	C–CO_2 balance, *NEP*					
European	–134	394	527	89	173	195
Siberian	370	463	93	418	35	8
Central Asian	218	465	247	366	93	25
Steppe regions of Russia on the whole	–134	465	599	231	202	87

x_{max}, maximum value; x_{min}, minimum value; A variation amplitude; $\bar{x}$ mean value; σ, standard deviation (in gC/(m^2 yr)); Cv, variation coefficient.

The calculations showed that heterotrophic respiration from soils of the natural steppe ecosystems of Russia cause the annual effluxe of 164 ± 37 MtC into the atmosphere. The heterotrophic emission from soils of the considered steppe ecosystems of the European steppe region is estimated as 102 ± 11 MtC/yr, or 62% of the annual R_H of the steppe regions of Russia. In cases of the Siberian and Central Asian regions, R_H values for the natural steppe ecosystems equal 31 ± 3 and 32 ± 3 MtC/yr, respectively, making up in each case about 19% of the annual soil heterotrophic respiration in steppe regions of Russia. According to the estimates, the annual heterotrophic respiration of natural steppe ecosystems of Russia is 60% of the annual NPP of these ecosystems.

Estimation of C–CO_2 Balance in Natural Steppe Ecosystems

Depending on the season, steppe ecosystems can act as a sink or source of atmospheric carbon. It should be noted that the intensity of production process in steppe ecosystems during the growing period determine the sign of annual carbon balance of these ecosystems. Instrumental measurements of carbon fluxes in ecosystems of the northern Kazakhstan showed that during the growing period natural steppes and abandoned lands accumulate atmospheric carbon (170 and 301 gC/m^2, respectively), while in the cold season these ecosystems demonstrate CO_2 emission with intensities of 128 and 155 gC/m^2, respectively [74]. The mixed-grass prairie (North Dakota, United States) in the warm season accumulates atmospheric carbon with intensity of 129 gC/m^2, while the intensity of carbon dioxide emission in the cold season is 84 gC/m^2 [75]. According to the calculations, in the warm season, natural steppes of different types; abandoned croplands of varied age; and abandoned pastures of the European, Siberian, and Central Asian regions of Russia accumulate 185 ± 168, 501 ± 58, and 438 ± 113 gC/m^2, respectively. In the cold season, steppe ecosystems of these regions emit 96 ± 21, 83 ± 21, and 72 ± 26 gC/m^2, respectively. The high carbon-accumulating ability of the natural steppes of the Siberian and Central Asian regions in the warm season is determined by the fact that these regions have 82 and 59% of the area, respectively (Fig. 3b), being occupied by secondary steppe ecosystems, i.e., ecosystems that intensively accumulate atmospheric carbon. It should also be noted that the vegetation cover of steppe ecosystems of the Siberian region is distinct in high productivity: NPP values are 1.5 times higher than productivity of steppes in the European region and 1.2 times higher than productivity of steppes in the Central Asian region (Table 4).

The annual carbon balance of steppe ecosystems, which are arid ecosystems, essentially depends on the total annual amount of precipitation and on its seasonal distribution. In the years when the regime of atmospheric precipitation was favorable for the development of grass stratum, natural steppe ecosystems can act as a significant sink of atmospheric carbon with NEP values ranging from 184 to 236 gC/(m^2 yr) [76–78]. In dry years, steppe ecosystems can both demonstrate carbon balance in a state close to the equilibrium one with the NEP values ranging from −32 to 45 gC/(m^2 yr) and be sources of CO_2 release into the atmosphere with NEP values ranging from −253 to −187 gC/(m^2 yr) [75, 76]. According to the calculations carried out, the most intensive sink of carbon (461–463 gC/(m^2 yr)) is characteristic of the natural ecosystems of meadow steppes of the Siberian region of Russia. The climatic conditions at which these ecosystems exist are favorable for the development of grass stratum during the growing period (NPP reaches 741–772 gC/(m^2 yr)). However, relatively harsh winters in this region considerably decelerate soil heterotrophic respiration in the cold season, with values from 50 to 120 gC/m^2. It was revealed that certain ecosystems of dry and desert steppes of the southeastern European region of Russia have negative NEP values in the range from −134 to −35 gC/(m^2 yr). These ecosystems exist under the climatic conditions unfavorable for development of the vegetation cover. The intensity of soil heterotrophic respiration in these ecosystems (R_H = 360–486 gC/(m^2 yr)) exceeds NPP (282–386 gC/(m^2 yr)).

The anthropogenic impact on the steppe ecosystems (grazing, plowing up, haying, fertilizing, and spring burns) have an essential effect on the carbon balance of these ecosystems. Instrumental measurements of carbon fluxes at the experimental sites of true steppe in North Kazakhstan [74] showed that the values of carbon balance in agricultural ecosystems (corn fields and hayfields) decrease in comparison with the sites of natural steppe ecosystems by 30–60% for the growing period and by 10–11% for the cold season. It follows from these results that the natural steppes of North Kazakhstan act as a sink of atmospheric carbon with an intensity of 42 gC/(m^2 yr), while corn fields and seeded hayfields of the same area are, vice versa, sources of atmospheric carbon with an intensity of 4–52 gC/(m^2 yr) [74]. The results of experimental measurements of carbon fluxes in the natural mixed-grass prairie and two types of pastures within its limits [75, 79] showed that the prairie sites free from anthropogenic impact are sinks of atmospheric carbon with intensity of 46 gC/(m^2 yr), whereas, at pasture sites within the mentioned prairie, the sink of carbon decreases by more than 20% (Table 5). Our calculations showed that abandoned pastures (they had been being under intensive grazing pressure and underwent considerable degradation) in dry steppes of the southeastern European region of Russia were sources of atmospheric CO_2 with NEP values ranging from −125 to −65 gC/(m^2 yr).

Abandoned croplands in the steppe zone, upon which steppe ecosystems are restoring, represent a significant sink of atmospheric carbon. According to [74], the intensity of the annual sink of atmospheric carbon into the abandoned ecosystem of true steppe is 3.5 times higher than the rate of carbon accumulation in the ecosystem of natural steppe (Table 5). It was noted in [78] that the sink of carbon into the natural steppe ecosystems of Khakassia did not exceed 130 gC/(m^2 yr), although abandoned croplands for 5 and 10 years accumulated 216 and 143 gC/(m^2 yr), respectively. Taking into account the statistical data about the distribution of different-type natural ecosystems within the steppe zone of Russia (Fig. 3), one can assume that about 50% of the annual sink of atmospheric carbon into these ecosystems is contributed by abandoned lands.

According to estimates, the mean NEP value for steppe ecosystems of Russia in general is 231 ± 202 gC/(m^2 yr), varying from 89 ± 173 gC/(m^2 yr) in

Table 5. Estimated $C–CO_2$ balance of steppe ecosystems in different regions of the world (based on instrumental data)

Location, steppe type, ecosystem(s), years of measurements	$C–CO_2$ balance, $gC/(m^2\ yr)$	Reference
Russia, Khakassia, true steppe:		[78]
natural steppe, 2002–2004	99–130	
5-year abandoned cropland, 2003–2004	216	
10-year abandoned cropland, 2004	143	
North Kazakhstan, true steppe:		[74]
natural steppe, 2002	42	
abandoned cropland, 2002	146	
crop field (spring wheat/barley), 2002	–52	
seeded hayfield, 2002	–4	
Mongolia, true steppe, pasture, 2002–2003	41	[90]
Hungary, dry steppe, natural steppe, 2003–2004	–80 to 180	[91]
United States, North Dakota, mixed-grass prairie:		
natural prairie, 1996–1999	45	[75]
pasture, 1996–1998	–19 to 52 (36)	[79]
seeded pasture, 1996–1998	–36 to 35 (–14)	[79]
United States, South Dakota, mixed-grass prairie, crop field (maize), 2008–2011	121–374 (230)	[92]
United States, Oklahoma, mixed-grass prairie:		
pasture, winter grazing, 1997	142	[93]
crop field (winter wheat), 2003–2007	–74 to 128 (37)	[92]
United States, Oklahoma, tallgrass prairie:		
natural prairie, 1997–1998	268	[94]
natural prairie, after spring burn, 1997	343	[93]
United States, Wisconsin, tallgrass prairie:		[25]
natural prairie, 2001–2004	–144 to 194 (38)	
65-year abandoned cropland, 2001–2004	–228 to 129 (–73)	
United States, Colorado, shortgrass prairie:		[76]
natural prairie, 2000–2003	–49 to 95 (3)	
natural prairie, 2004–2007	–187 to 78 (–69)	
pasture, moderate continuous grazing, 2004–2006	–32 to 128 (73)	
pasture, high continuous grazing, 2004–2006	–26 to 57 (29)	
pasture, moderate spring/fall grazing, 2004–2006	–102 to –61 (–85)	
United States, Illinois, tallgrass prairie, crop field (maize), 1997, 1999, 2001, 2003, 2006	295–575 (412)	[92]
United States, Texas, shortgrass steppe, pasture, 2010–2011	–353 to 70	[76]
United States, Wyoming, shortgrass steppe, natural steppe, 1997–1998	39 to 109	[76]
Canada, Alberta, mixed-grass prairie, crop field (spring wheat), 2007	–37	[92]
Canada, Manitoba, tallgrass prairie, crop field (spring wheat), 2008	229	[92]

Positive values of the balance mean that the ecosystem is a sink of atmospheric carbon; negative values mean the ecosystem is a source of atmospheric carbon; mean values of $C–CO_2$ balance for its variation range are in parentheses.

the European steppe region to 418 ± 35 $gC/(m^2\ yr)$ in the Siberian one (Table 3). The natural steppe ecosystems in the European region of Russia are characteristic of the largest amplitude of NEP variation (527 $gC/(m^2\ yr)$), because there are not only ecosystems that are intensive sinks of atmospheric carbon (more than 250 $gC/(m^2\ yr)$), but also ecosystems acting as sources of atmospheric carbon of moderate intensity (125–134 $gC/(m^2\ yr)$). The estimate of $C–CO_2$ balance in the natural steppe ecosystems exceeds the mean sink of carbon dioxide in forest areas of Russia about 3.5 times, estimated at 66 ± 15 $gC/(m^2\ yr)$ [59]; it is also 5 times higher than the sink of carbon in boreal forests of the Northern Hemisphere, which is estimated at 46 gC/(m^2 yr) [80], and 2.9–5.6 times higher than the sink of atmospheric carbon into the subarctic tundra ecosystems of the northeastern European part of Russia, estimated at 41 to 79 gC/(m^2 yr) [81].

The calculations made have shown that the natural steppe ecosystems of Russia annually accumulate 111 ± 97 MtC. The obtained value of atmospheric carbon sink into natural steppe ecosystems of Russia is consistent with the variation range of this parameter, 92 to 121 MtC/yr, as published in [66]. Sinks of atmospheric carbon into the natural and secondary steppe ecosystems of the European, Siberian, and Central Asian regions of Russia are 20, 37, and 43%, respectively, of the annual accumulation of carbon by these ecosystems and 18, 57, and 60%, respectively, of the annual NPP of these ecosystems.

The study results have shown that the natural ecosystems of the Russian steppe zone demonstrate a high potential of atmospheric carbon sink and the disturbance of carbon cycle balance in these ecosystems. One of the causes of carbon imbalance in the considered ecosystems is that about 58% of the area of natural steppes is occupied by abandoned croplands and former pastures (Fig. 3b). The restoration of species composition of vegetation is continuing, as is the restoration of soil cover, back to the patterns typical of natural steppe ecosystems. At all stages of restoration, the productivity of vegetation communities and contents of soil organic matter considerably increase in secondary steppe ecosystems [37, 52, 82, 83]. When restoring steppe ecosystems, an increase in intensity of soil heterotrophic respiration is reported; however, the significant growth of R_H in these ecosystems is controlled by the aridity of climate and growth of thermal insulation of the upper soil layers during the accumulation of the above-soil detrital layer and the recharge of organic matter in steppe soils [84].

CONCLUSIONS

The steppe zone occupies 10% of the territory of Russia. The steppe ecosystems of Russia are subdivided into meadow, true, dry, and desert ones, which occupy 41, 27, 19, and 13%, respectively, of the studied steppe area. Natural and secondary steppes in the territory of Russia occupy 500 000 km^2, making up 28% of the total area of the steppe zone in the Russian territory and 3% of the area of all Russian terrestrial ecosystems. The considered ecosystems are scattered in the form of separated steppe massifs in the southern parts of Russia.

According to the land inventory of the steppe territories, the European region of Russia concentrates 65% of the remaining steppe ecosystems of Russia; the Siberian and Central Asian regions concentrate 9 and 26%, respectively. Our analysis of the data on the state and use of lands in Russia has shown that the present-day areas occupied by secondary steppe ecosystems exceed the areas of natural steppe ecosystems 1.4 times, which was revealed during land inventory of steppe territories. In the European region of Russia, the area of secondary steppe ecosystems is 1.1 times less than the area covered by natural steppes. In the Siberian and Central Asian regions, the areas of secondary steppe ecosystems exceed the areas of natural steppes 4.5 and 1.4 times, respectively.

Our calculations have shown that natural steppe ecosystems of Russia presently act as a significant sink of atmospheric CO_2. These ecosystems annually accumulate 111 ± 97 MtC (407 ± 355 $MtCO_2$). Sink of carbon into natural steppe ecosystems is 40% of the production component of the carbon cycle of these ecosystems. It should be noted that carbon accumulated by steppe ecosystems is accumulated mainly in the belowground parts of plants, in phytodetritus (chiefly belowground), and in soil humic substances.

The amount of atmospheric carbon, annually accumulated by the considered steppe ecosystems, is comparable to the sink of carbon into managed forests of Russia (96 MtC/yr [85]), making up about 20% of the contemporary sink of carbon into forest ecosystems of Russia, which is estimated at 530–595 MtC/yr [86].

Terrestrial ecosystems of Russia annually accumulate 0.6–1.3 Gt of atmospheric carbon [14, 87–89]. According to our calculations made in the framework of the present study, natural and secondary steppe ecosystems of Russian Federation provide from 8 to 19% of atmospheric carbon sink into terrestrial ecosystems of Russia, making up a significant part of the annual carbon balance of the Russian territory.

Our study has shown that steppe regions possess a high carbon-accumulating potential. The estimates show that natural steppe ecosystems can markedly affect the consequences of climate change and act as essential climate-stabilizing systems. Anthropogenic impact considerably changes the intensity of production- and destruction-related carbon fluxes into steppe ecosystems and, thus, have a significant influence on carbon gas exchange between steppe territories and atmosphere.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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